Enter the world of artificial intelligence by working with global AI giants and disruptors.

We're on the lookout for a Sr. LLM Engineer - with strong backend expertise.

As an LLM leader at Acme AI, you'll play a critical role in architecting and implementing backend systems and LLM integrations for powering cloud-based API-driven Retrieval Augmented Generation (RAG) and NL2SQL applications for our international clients.

KEY RESPONSIBILITIES

General responsibilities include:

- Design and develop Python-based backend services (Fast API / Flask) for LLM based RAG systems and data engines
- Build and optimise document ingestion, embedding, retrieval and generation pipelines using vector databases (PG Vector, FAISS, Pinecone)
- Develop and optimise RAG workflows, including document chunking, context retrieval, and response generation
- Integrate and orchestrate LLM providers (OpenAI, Anthropic, Azure OpenAI) through LangChain or custom orchestration layers
- Build and optimise APIs for documentation, retrieval and generation
- Implement NL2SQL pipelines (baseline and agentic) for natural-language-to-query conversion and visualisation suggestion
- Ensure scalability, reliability, and security of Python services deployed on AWS (ECS, Lambda, S3, RDS) and MS Azure
- Ensure data privacy, security and compliance across backend systems
- Monitor performance, latency, and cost of LLM calls, embeddings, and retrieval operations
- Collaborate closely with the Controller (Node.js) team to align orchestration and response pipelines
- Lead code reviews, unit testing, and CI/CD automation for Python-based backend components
- Document APIs and backend workflows, ensuring clarity for frontend and data engineering teams

EXPERIENCE + QUALIFICATION

- 5–7 years of backend engineering experience with Python (FastAPI, Flask, or Django)
- Proven experience building LLM-integrated RAG systems or AI-driven backend architecture
- Strong understanding of vector databases (PGVector, FAISS, Pinecone, Weaviate) and embedding models
- Hands-on experience with LangChain, LlamaIndex, or similar LLM orchestration frameworks
- Strong knowledge of AWS services (ECS, Lambda, S3, RDS, CloudWatch) and CI/CD pipelines
- Familiarity with prompt engineering, token optimisation and multi-model LLM orchestration
- Knowledge of data processing workflows (CSV/XLS ingestion, schema parsing, SQL query generation)

- Commitment to best practices in code quality, security, and cloud-native design

NICE TO HAVE

- Experience with agentic LLM pipelines (multi-step reasoning and tool use)
- Familiarity with multi-tenant architectures and event-driven systems (SQS, SNS, Kafka)
- Knowledge of MLOps / LLMOps practices for monitoring and fine-tuning model usage
- Background in data visualisation APIs or analytics-driven query generation
- Experience in Microsoft Azure services and technologies

WORK DETAILS

This is a full-time position, where you'll be working in a supportive, fast-paced environment with access to learning and development opportunities.

- Type: Full-time, onsite with some remote flexibility
- Schedule: Standard 8 hours, five-days engagement under contract
- Location: Acme AI Ltd, Level 5, House 385, Road 6, Mirpur DOHS, Dhaka 1216
- Salary: BDT 80,000-130,000 (offered based on experience and test results)

BENEFITS

- Competitive salary based on experience
- Festival bonuses
- Access to company health fund (upon eligibility; since health and hospitalisation insurance sucks!)
- Quarterly profit-sharing (upon eligibility)
- Acme AI is the largest AI enterprise in Bangladesh - having the largest AI data curation team in the country. Therefore, your developed LLM solutions can benefit from on-demand SFT/RLHF support subject to client approval.